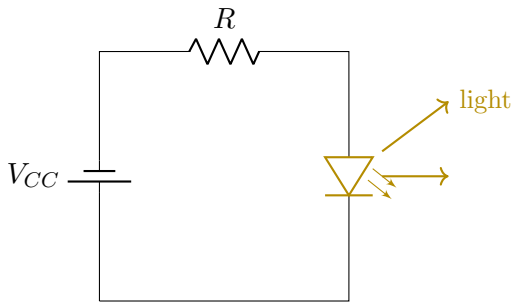
Designing with LEDs

A Practical Study Guide

Selecting components, calculating resistors,
and matching circuits to design goals



Topics covered:
LED physics, forward voltage, current limiting
Ohm's law applied to LED circuits
Selecting operating current (Low / Medium / High)
Resistor power ratings and package sizing
Matching LEDs to voltage and current budgets
Interpreting datasheets and handling uncertainty

Exercises and worked examples throughout. Answer key in Appendix A.

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1 What Is an LED?

A **Light Emitting Diode (LED)** is a semiconductor device that emits light when current flows through it in the forward direction. Unlike a resistor, whose resistance is fixed by Ohm's law, an LED presents a nearly constant *voltage drop* when conducting, and is very sensitive to the current flowing through it.

- **Forward Voltage (V_f):** The voltage drop across the LED when it is conducting. This is fixed by the LED's chemistry and color, not by the circuit. Typical values range from about 1.8 V (red) to 4.0 V (violet/UV).
- **Forward Current (I_{LED}):** The current flowing through the LED. Brightness is approximately proportional to current.
- **Maximum Current (I_{max}):** The absolute maximum current the LED can handle continuously without damage. Never exceed this.
- **Reverse breakdown:** LEDs are destroyed if significant voltage is applied backwards. Always connect the anode (+) toward the positive supply.

1.1 The LED's I–V Curve and the Load Line

Unlike a resistor, the LED does *not* obey Ohm's law directly. Its current-versus-voltage relationship is **highly nonlinear**: below V_f almost no current flows; above it, current rises exponentially and extremely steeply. This behaviour is described by the *diode equation*, but for practical circuit design the key points are:

- The curve is nearly **flat near zero** — the LED is essentially an open circuit below its threshold voltage.
- The curve becomes **nearly vertical above** V_f — once conducting, a tiny increase in voltage produces a huge increase in current.
- The steepness means that connecting an LED directly to a fixed voltage source is dangerous: with no resistance to limit current, the LED sits on the vertical part of its curve and draws as much current as the source can supply, burning out in milliseconds.

The Load Line

The series resistor R imposes its own constraint on the circuit, described by a straight line called the **load line**. By Kirchhoff's voltage law, whatever voltage the LED takes (V_{LED}), the resistor gets the rest:

$$V_R = V_{CC} - V_{LED}$$

and the current through the resistor is $I = V_R/R$, so:

$$I = \frac{V_{CC} - V_{LED}}{R}$$

This is a straight line on the I–V axes. It has two easy-to-find endpoints:

- **X-intercept** (current = 0): $V_{LED} = V_{CC}$. If no current flows, the full supply appears across the LED.

- **Y-intercept** (voltage across LED = 0): $I = V_{CC}/R$. If the LED were a short circuit, the resistor would limit current to V_{CC}/R.

The Operating Point

The LED and resistor are in the same series loop, so they must simultaneously satisfy *both* constraints — the LED's I–V curve *and* the load line. The only point that satisfies both at once is where the two lines **intersect**. This intersection is the **operating point**: the actual current and voltage at which the circuit runs.

Graphically, you draw the LED curve and the load line on the same axes. The operating point falls naturally on the steep part of the LED curve, in the region where the curve is nearly vertical. This has an important consequence: the steep curve means the operating point's *current* is mostly controlled by the load line (i.e. the resistor), while the *voltage* is pinned near V_f by the LED. This is exactly the behaviour we want — the resistor sets the current, and the LED self-selects its voltage.

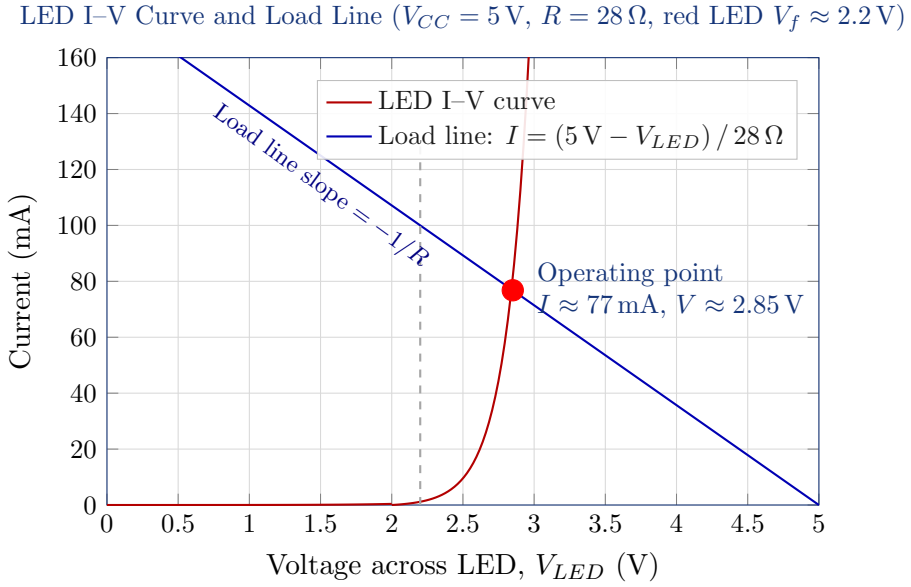


Figure 1: The LED I–V curve (red) and the load line (blue) for a 28 Ω resistor on a 5 V supply. The load line runs from (0, V_{CC}/R) on the current axis to (V_{CC}, 0) on the voltage axis. The operating point (filled circle) is the unique intersection: the only combination of current and voltage that satisfies both the LED's physics and Ohm's law in the resistor simultaneously. Note that the operating point lies to the right of V_f on the steep part of the curve, confirming that the LED is conducting. Moving to a smaller R rotates the load line upward (steeper slope), raising both the current and the operating-point voltage slightly; a larger R rotates it downward.

Tip

Reading the load line intuitively. The load line's slope is $-1/R$: a smaller resistor gives a steeper line and higher current; a larger resistor gives a shallower line and lower current. Changing V_{CC} shifts both intercepts, sliding the line along both axes. In practice you never need to draw this graph — the equations in Section 5 give the same answer analytically — but understanding it explains *why* the resistor controls current rather than voltage, and why

the LED's voltage barely changes even when you change the resistor.

Common Mistake

Do **not** connect an LED directly to a voltage source. Without a resistor there is no load line — or equivalently, the load line is vertical (zero resistance), and the intersection with the LED curve can be anywhere up the steep part of the curve, determined only by how much current the source can supply. The LED will burn out in milliseconds.

2 The Basic LED Circuit

The solution is simple: add a **series resistor** R between the supply and the LED. The resistor absorbs the difference between the supply voltage and V_f, and limits the current to a safe value.

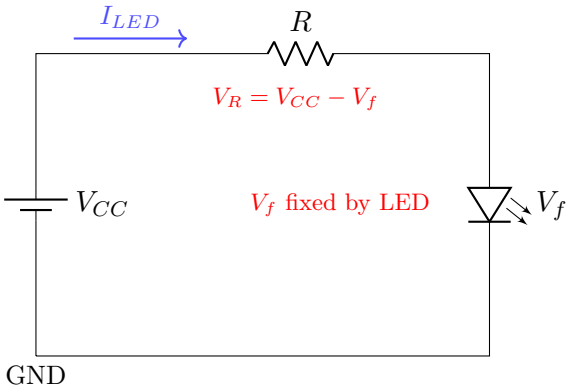


Figure 2: The fundamental LED circuit. The resistor R limits current by dropping the voltage difference $V_R = V_{CC} - V_f$.

2.1 The Governing Equations

The three equations you need are all forms of Ohm's law.

The Three LED Circuit Equations

$$V_R = V_{CC} - V_f \tag{1}$$

$$I_{LED} = \frac{V_R}{R} = \frac{V_{CC} - V_f}{R} \tag{2}$$

$$R = \frac{V_{CC} - V_f}{I_{LED}} \tag{3}$$

where:

- V_{CC} = supply voltage (V)
- V_f = LED forward voltage (V) — from the datasheet
- V_R = voltage across the resistor (V)
- I_{LED} = LED current (A or mA) — you choose this

- R = resistor value (Ω)

Tip

The LED current I_{LED} is the *same* as the current through the resistor — they are in series, so only one current flows in the whole loop.

2.2 Worked Example: Finding R

Given:

- Supply voltage: $V_{CC} = 5.0\,\text{V}$
- Red LED: $V_f = 2.2\,\text{V}$ (typical), $I_{max} = 250\,\text{mA}$
- Desired current: $I_{LED} = 100\,\text{mA}$

Step 1: Find the voltage across the resistor.

$$V_R = V_{CC} - V_f = 5.0 - 2.2 = 2.8\,\text{V}$$

Step 2: Calculate the required resistance.

$$R = \frac{V_R}{I_{LED}} = \frac{2.8\,\text{V}}{0.100\,\text{A}} = 28\,\Omega$$

Step 3: Choose the nearest standard (E24) value.
The nearest E24 value below $28\,\Omega$ is **27** Ω. The nearest above is **30** Ω.
With $27\,\Omega$: $I_{LED} = 2.8/27 = 103.7\,\text{mA}$ (slightly over target, but under I_{max})
With $30\,\Omega$: $I_{LED} = 2.8/30 = 93.3\,\text{mA}$ (slightly under target)
Either is acceptable. Choose $27\,\Omega$ for slightly more brightness, $30\,\Omega$ for slightly more headroom below I_{max}.

Q1. A blue LED has $V_f = 2.95\,\text{V}$ and is powered from a 5 V supply. You want $I_{LED} = 80\,\text{mA}$. What resistor do you need?

$V_R =$ _____ $R_{calc} =$ _____ Nearest E24 = _____

(See Appendix A for the answer.)

3 Understanding LED Specifications

Before you can design a circuit, you need to read and understand the LED datasheet. This section explains the most important specifications.

3.1 Forward Voltage (V_f): Typical, Minimum, and Maximum

Datasheets usually give three values for V_f:

Value	Meaning	How to use it
V_f^{typ}	Most LEDs from this batch	Use for nominal calculations
V_f^{min}	Best-case (lowest voltage, highest current for given R)	Use to find maximum possible current
V_f^{max}	Worst-case (highest voltage, lowest current for given R)	Use to verify the LED still lights at minimum brightness

Common Mistake

Using only V_f^{typ} can mislead you. If V_f^{max} is much higher than V_f^{typ}, some LEDs from the production run will be noticeably dimmer than expected. **Example:** The violet LED (row 1015 in our project) has $V_f^{typ} = 2.8\,\text{V}$ but $V_f^{max} = 4.0\,\text{V}$. With a $120\,\Omega$ resistor on a 5 V supply:

- At V_f^{typ}: $I = (5.0 - 2.8)/120 = 18.3\,\text{mA}$
- At V_f^{max}: $I = (5.0 - 4.0)/120 = 8.3\,\text{mA}$

The dimmest LEDs produce less than half the current of the brightest. Always check the full V_f range.

3.2 Maximum Current (I_{max})

I_{max} is the *absolute* maximum continuous forward current. Exceeding it will destroy the LED by overheating its junction. It is **not** a target — it is a hard ceiling.

Tip

A good operating current is **50–75% of** I_{max}. This gives good brightness while keeping the junction temperature low, extending LED lifetime and maintaining spectral stability.

3.3 Power Dissipation (P_{max})

Every LED also has a maximum power dissipation rating:

$$P_{LED} = V_f \times I_{LED}$$

This must stay below the datasheet's P_{max}. For the violet LED: $P_{max} = 120\,\text{mW}$. At $V_f = 2.8\,\text{V}$ and $I = 18\,\text{mA}$: $P = 2.8 \times 0.018 = 50.4\,\text{mW}$. Well within limit.

Quick Check 3.1

Q2. A green LED has $V_f = 2.8\,\text{V}$, $I_{max} = 240\,\text{mA}$, and $P_{max} = 800\,\text{mW}$. You plan to run it at $I = 150\,\text{mA}$.

(a) What is the power in the LED?
(b) Is this within the power limit?
(c) What percentage of I_{max} are you using?

(See Appendix A.)

3.4 Luminous Output

LEDs may specify brightness in several units depending on their class:

- **lm (lumens):** Total light output. Used for power/high-brightness LEDs.
- **mcd (millicandela):** Intensity in a specific direction. Used for indicator LEDs.
- **mW (milliwatts):** Optical radiant power. Used for UV/IR LEDs where human visibility is less meaningful.

Brightness is *approximately proportional to current*. Running at 50% of I_{max} gives roughly 50% of the rated luminous output.

4 Choosing an Operating Current

One of the most important design decisions is how hard to drive your LED. This section introduces a systematic approach using three defined levels.

4.1 The Three Brightness Levels

Rather than picking current arbitrarily, define three levels as fractions of I_{max}:

Level	% of I_{max}	When to use
Low	≈40%	Calibration, testing, or when camera exposure can compensate
Medium	≈60%	Normal operation; good brightness with comfortable thermal margin
High	≈80%	Maximum brightness needed; monitor temperature; shorter LED lifetime

Given: Green LED, $I_{max} = 240\,\text{mA}$.

$$I_{Low} = 0.40 \times 240 = 96\,\text{mA}$$
$$I_{Medium} = 0.60 \times 240 = 144\,\text{mA}$$
$$I_{High} = 0.80 \times 240 = 192\,\text{mA}$$

Before accepting these, check the power budget. If four LEDs run in parallel, total current at Medium $= 4 \times 144 = 576\,\text{mA}$. If the budget is 500 mA, the Medium target must be reduced:

$$I_{Medium,max} = \frac{500\,\text{mA}}{4\text{ LEDs}} = 125\,\text{mA} \quad (52\%\text{ of } I_{max})$$

This is still a good operating point, just slightly below the 60% guideline.

4.2 The Power Budget Constraint

When multiple LEDs run in parallel, the total current is:

$$I_{total} = N \times I_{LED} \tag{4}$$

This must not exceed your supply's current limit. Always check I_{total} before finalizing a resistor value.

Q3. You have a red LED with $I_{max} = 250\,\text{mA}$. You want to run 4 LEDs in parallel from a supply limited to 500 mA.
(a) What is I_{Medium} per LED at 60% of I_{max}?
(b) What is the total current at Medium? Does it fit the budget?
(c) What is the maximum per-LED current that fits the 500 mAbudget?
(d) What percentage of I_{max} is that maximum?
(See Appendix A.)

5 Calculating the Current-Limiting Resistor

5.1 Step-by-Step Procedure

1. Choose the desired LED current I_{LED} (using the brightness-level approach from Section 4, constrained by the power budget).
2. Compute the voltage across the resistor:
$$V_R = V_{CC} - V_f^{typ}$$
3. Calculate the exact resistor value:
$$R_{calc} = \frac{V_R}{I_{LED}}$$
(Use amperes, not milliamperes, in this formula.)
4. Round to the nearest **E24 standard value** (see Table 1).
5. Calculate the *actual* current with the standard resistor:
$$I_{LED}^{actual} = \frac{V_R}{R_{E24}}$$
6. Verify: $I_{LED}^{actual} < I_{max}$ and $N \times I_{LED}^{actual} <$ budget.

5.2 The E24 Resistor Series

Standard resistors come in the E24 series, which provides 24 values per decade. These are the most commonly available values.

Table 1: E24 Standard Resistor Values (multiply by any power of 10)

1.0	1.1	1.2	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.7	3.0
3.3	3.6	3.9	4.3	4.7	5.1	5.6	6.2	6.8	7.5	8.2	9.1

Tip

To find the nearest E24 value to any calculated resistance: divide by the appropriate power of 10 to get a value between 1 and 10, find the nearest entry in Table 1, then multiply back. For example, $R_{calc} = 18.7\,\Omega$: divide by 10 $\rightarrow$ 1.87; nearest E24 is 1.8; multiply by 10 $\rightarrow$ **18** Ω.

5.3 Full Worked Example

Given:

- Supply: $V_{CC} = 5.0\,\text{V}$
- LED: $V_f^{typ} = 2.80\,\text{V}$, $I_{max} = 240\,\text{mA}$
- 4 LEDs in parallel
- Current budget: 500 mA
- Target level: Medium (60% of I_{max})

Step 1: Initial target current.

$$I_{LED}^{target} = 0.60 \times 240 = 144\,\text{mA}$$

Step 2: Check power budget.

$$I_{total} = 4 \times 144 = 576\,\text{mA} \quad > 500\,\text{mA} \quad \text{(over budget!)}$$

Reduce: $I_{LED}^{max} = 500/4 = 125\,\text{mA}$. Use $I_{LED} = 122\,\text{mA}$(51% of I_{max}).

Step 3: Voltage across resistor.

$$V_R = 5.0 - 2.80 = 2.20\,\text{V}$$

Step 4: Calculate R_{calc}.

$$R_{calc} = \frac{2.20}{0.122} = 18.03\,\Omega$$

Step 5: Nearest E24 value. 18.03 Ω $\rightarrow$ divide by 10 $\rightarrow$ 1.803 $\rightarrow$ nearest E24 is 1.8 $\rightarrow$ multiply by 10 $\rightarrow$ **18** Ω.

Step 6: Actual current.

$$I_{LED}^{actual} = \frac{2.20}{18} = 0.1222\,\text{A} = 122.2\,\text{mA}$$

Step 7: Verify.

- 122.2 mA $<$ 240 mA (I_{max}) ✓
- $4 \times 122.2 = 488.9\,\text{mA} < 500\,\text{mA}$ (budget) ✓
- 122.2/240 = 50.9% of I_{max} ✓

6 Resistor Power Rating

A resistor carrying current dissipates power as heat. If this exceeds the resistor's power rating, it will overheat and drift in value (or fail). Always calculate resistor power and select an appropriate package.

6.1 Calculating Resistor Power

$$P_R = I_{LED}^2 \times R \tag{5}$$

From Example 5.1: $I_{LED} = 122.2\,\text{mA}$, $R = 18\,\Omega$.

$$P_R = (0.1222)^2 \times 18 = 0.01493 \times 18 = 0.269\,\text{W} = 269\,\text{mW}$$

6.2 Choosing the Right Package

SMD resistors come in standard packages with different power ratings:

Package	Size (mm)	Power Rating	Notes
0402	1.0×0.5	0.063 W	Indicator LEDs (<63 mWin resistor)
0603	1.6×0.8	0.100 W	Low-power applications
0805	2.0×1.25	0.125 W	Moderate current
1206	3.2×1.6	0.25 W	**Recommended for power LED resistors**
2512	6.3×3.2	1.0 W	High-power applications

Tip

Always use a package rated for **at least twice** the calculated dissipation. This "2× derating" keeps the resistor cool and extends its life. In Example 6.1, $P_R = 269\,\text{mW}$: a 1206 rated 0.25 W is marginal; consider a 2512 (1 W) for a comfortable margin.

Common Mistake

The most common beginner mistake in LED driver design is using a 0402 or 0603 resistor for a power LED circuit. At 100 mAor more, the resistor can reach temperatures that change its value, cause solder joint failures, or start a fire on a PCB. Calculate P_R for *every* design.

Quick Check 6.1

Q4. Calculate the power dissipated in the resistor for each case:
(a) Blue LED: $R = 18\,\Omega$, $I_{LED} = 114\,\text{mA}$. What is P_R? What package do you need?
(b) Violet LED: $R = 120\,\Omega$, $I_{LED} = 18\,\text{mA}$. What is P_R? What package do you need?
(c) Red LED: $R = 24\,\Omega$, $I_{LED} = 117\,\text{mA}$. What is P_R? What package do you need?

(See Appendix A.)

7 Parallel LED Groups

7.1 Why Parallel, Not Series?

When illuminating a surface evenly it is tempting to wire all LEDs in a single series string with one shared resistor. This section explains in detail why that approach fails when simple resistors are used, and what the correct alternative is.

7.1.1 How a Series String Works in Principle

In a series string every LED carries exactly the same current — there is only one path for current to flow. The resistor is calculated for the total string voltage:

$$R = \frac{V_{CC} - N \cdot V_f}{I_{LED}}$$

For example, 3 red LEDs ($V_f = 2.2\,\text{V}$) on a 12 V supply at 100 mA:

$$R = \frac{12 - (3 \times 2.2)}{0.100} = \frac{12 - 6.6}{0.100} = 54\,\Omega$$

This looks straightforward. In fact, series strings *are* the standard approach in commercial LED lighting — but only when paired with a **constant-current driver IC** that actively regulates the current regardless of what the LEDs do. The problems described below are specific to using a simple resistor instead of a driver IC.

7.1.2 Problem 1: A Resistor Cannot Stabilize a Series String on 5 V

The fundamental role of the resistor is to absorb the difference between the supply and the LED voltage, providing a stable current. This only works when the voltage *across the resistor* is large enough relative to the LED voltage variations.

With 3 red LEDs in series on a 5 V supply:

$$V_R = 5.0 - (3 \times 2.2) = 5.0 - 6.6 = -1.6\,\text{V}$$

The math is negative — **the supply voltage is not high enough for 3 red LEDs in series at all**. Only 2 red LEDs fit: $V_R = 5.0 - 4.4 = 0.6\,\text{V}$, which is barely enough headroom for a resistor to function (recall the 1 V headroom rule from Section 8). For blue or green LEDs ($V_f \approx 2.8$–$3.0\,\text{V}$), even 2 in series exceeds 5 V. A 5 V supply can only drive series strings of *one or two* LEDs of any color — making it impractical for groups of 4 or 6.

7.1.3 Problem 2: Vf Production Spread Destabilizes the Whole String

Every LED from a production batch has a slightly different V_f, typically varying ±0.1–0.3 V around the nominal value. In a series string, these variations *add up* across all LEDs. With 4 LEDs each having a ±0.2 V spread, the total string voltage could range from $4 \times (2.0) = 8.0\,\text{V}$ to $4 \times (2.4) = 9.6\,\text{V}$

— a 1.6 V range. The resistor, which must absorb V_{CC} minus the string voltage, sees its share swing widely, causing a proportional swing in current.

Because the LED I-V curve is very steep near V_f, even a small change in the *net* voltage affects the current dramatically. The resistor provides damping only in proportion to how large V_R is compared to the total string voltage. On a 5 V supply with most of the voltage consumed by LEDs, V_R is so small that the resistor provides almost no stabilization at all.

Two red LEDs, $V_f^{nom} = 2.2\,\text{V}$. LED A has $V_f = 2.1\,\text{V}$; LED B has $V_f = 2.3\,\text{V}$. Supply: 5 V. **Series, $R = 30\,\Omega$ (designed for nominal):**

$$I = \frac{5.0 - (2.1 + 2.3)}{30} = \frac{5.0 - 4.4}{30} = \frac{0.6}{30} = 20\,\text{mA}$$

Now suppose both LEDs happen to be at $V_f^{max} = 2.3\,\text{V}$:

$$I = \frac{5.0 - 4.6}{30} = \frac{0.4}{30} = 13\,\text{mA} \quad (35\%\text{ less current})$$

A 0.2 V Vf shift per LED cut the current by a third because V_R is tiny.
Parallel, $R = 14\,\Omega$ each (designed for nominal 100 mA): LED A: $I_A = (5.0 - 2.1)/14 = 2.9/14 = 207\,\text{mA}$— *wait, that exceeds I_{max}!*
Lesson: Parallel also has issues if V_R is too small. Using a 5 V supply with a proper $V_R \geq 1\,\text{V}$: $R = (5.0 - 2.2)/0.100 = 28\,\Omega$. LED A ($V_f = 2.1\,\text{V}$): $I_A = 2.9/28 = 103.6\,\text{mA}$. LED B ($V_f = 2.3\,\text{V}$): $I_B = 2.7/28 = 96.4\,\text{mA}$. Only a 7% variation — each LED self-regulates independently.

7.1.4 Problem 3: Different Colors Cannot Share a Series String

Different LED colors have very different V_f values:

Color	Typical V_f
Red	2.0–2.4 V
Green	2.6–3.2 V
Blue	2.8–3.2 V
Violet	2.8–4.0 V

To mix a red and a blue LED in a series string on 5 V:

$$V_{string} = 2.2 + 2.95 = 5.15\,\text{V} \quad > 5.0\,\text{V}$$

The combined forward voltage already exceeds the supply. No current would flow at all — or if the supply voltage is slightly above nominal the LED with the lower breakdown margin could be damaged. Mixed-color series strings on 5 V are simply impossible by arithmetic.

7.1.5 Problem 4: One Failure Kills the Whole Group

When an LED fails, it most commonly fails **open circuit** (the bond wire inside breaks). In a series string, one open LED breaks the only current path and extinguishes all the others instantly. In a

parallel group with individual resistors, a failed LED removes only one branch — the remaining LEDs continue operating, and the failure is immediately visible in the image as one dark spot rather than total illumination loss.

7.1.6 Problem 5: You Cannot Adjust LEDs Independently

In a parallel group, changing one LED's individual resistor adjusts *that LED only*. This is useful during calibration. In a series string, the single resistor controls all LEDs together — there is no way to compensate for one particularly dim or bright LED without affecting the rest.

7.1.7 When Series Strings Are the Right Choice

It is worth understanding that commercial LED products *do* use series strings extensively — but with a constant-current driver IC, not a resistor. The IC continuously measures the actual current and adjusts its output voltage to keep it exactly at the target, regardless of V_f spread, temperature, or number of LEDs. This makes series strings efficient (no resistor heat loss) and tightly controlled.

For our circuit, which uses simple resistors and a fixed 5 V supply, parallel LEDs with individual resistors is the only practical approach. The resistor approach trades some efficiency for simplicity, and works well as long as $V_R \geq 1\,\text{V}$.

7.1.8 Summary of Series vs. Parallel (with Resistor)

Property	**Series + 1 resistor**	**Parallel + individual R**
Works on 5 V with multiple LEDs	Rarely	Yes
Sensitive to V_f production spread	Very (cumulative)	Low (independent)
Supports mixed colors	No	Yes
Failure of one LED	All go dark	Others continue
Independent per-LED adjustment	No	Yes
Suitable for resistor-based design	**No**	**Yes**

Individual resistors with parallel LEDs is therefore the correct topology for this design. Each LED is independently current-controlled regardless of its individual V_f, the 5 V supply is fully compatible, colors can be freely mixed, and individual LEDs can be trimmed or replaced without disturbing their neighbours.

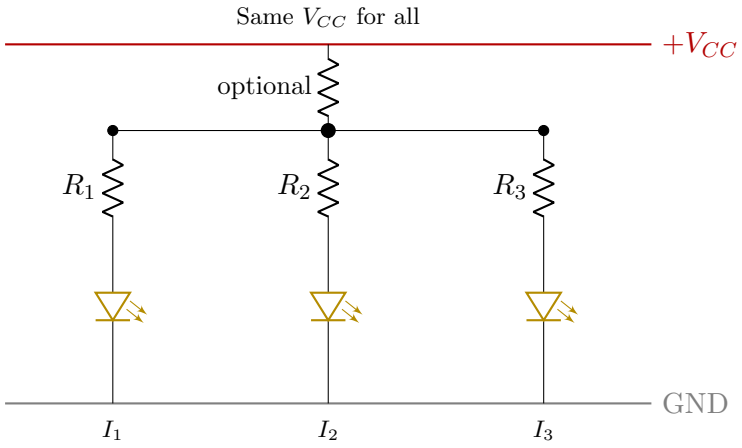


Figure 3: Three parallel LEDs, each with its own current-setting resistor R_1, R_2, R_3. An optional shared series resistor R_{sh} trims the whole group simultaneously.

7.2 The Optional Shared Resistor

A single **shared series resistor** R_{sh} can be added between the supply and the distribution bus to trim the brightness of the entire group without changing individual resistors. It carries the full group current:

$$I_{sh} = N \times I_{LED}$$

Common Mistake

R_{sh} carries *all* the group current. At $N = 4$ and $I_{LED} = 125\,\text{mA}$, $I_{sh} = 500\,\text{mA}$. A standard 1/4 W resistor rated for 250 mAwould fail. Always use a **power-rated wirewound resistor** ($\geq$2 W) for R_{sh}, or replace it with a wire (0 Ω) and use software exposure time to tune brightness.

Suppose 4 red LEDs ($V_f = 2.2\,\text{V}$) each have R_1–$R_4 = 22\,\Omega$ on a 5.0 V supply. Without R_{sh} ($R_{sh} = 0$):

$$I_{LED} = \frac{5.0 - 2.2}{22} = \frac{2.8}{22} = 127\,\text{mA}$$

Add $R_{sh} = 10\,\Omega$. The total resistance in series with each LED becomes:

$$R_{total} = R_i + N \times R_{sh}/N$$

Wait — R_{sh} is in series with the whole group, not each LED individually. The current per LED is found by:

$$I_{LED} = \frac{V_{CC} - V_f}{R_i + R_{sh}} = \frac{5.0 - 2.2}{22 + 10} = \frac{2.8}{32} = 87.5\,\text{mA}$$

Adding just 10 Ω of shared resistance reduced current from 127 mAto 87.5 mA(a 31% reduction in brightness) — a powerful and easy group trim.

Q5. Four blue LEDs ($V_f = 2.95\,\text{V}$, $R_i = 18\,\Omega$ each) are on a 5.0 V supply.
(a) What is I_{LED} with no shared resistor?
(b) What is I_{total}?
(c) You add $R_{sh} = 5\,\Omega$. What is the new I_{LED}?
(d) What power does R_{sh} dissipate at the current from part (b)?
(See Appendix A.)

8 Supply Voltage Considerations

8.1 Headroom: Why Supply Voltage Matters

For stable current control, the voltage across the resistor ($V_R = V_{CC} - V_f$) should be **at least 1 V**. If V_R is too small, tiny variations in V_f between LEDs or with temperature cause large swings in current.

A blue LED with $V_f = 2.95\,\text{V}$ on a 3.3 V supply:

$$V_R = 3.3 - 2.95 = 0.35\,\text{V}$$

With $R = 10\,\Omega$: $I_{LED} = 0.35/10 = 35\,\text{mA}$.
But if this particular LED has $V_f = 3.05\,\text{V}$ (within production spread):

$$I_{LED} = (3.3 - 3.05)/10 = 0.025\,\text{A} = 25\,\text{mA}$$

That's a 29% current variation from a 0.1 V V_f change. The LED group will have noticeable brightness differences.
On a 5 V supply with $R = 22\,\Omega$:

$$I_{LED} = (5.0 - 2.95)/22 = 93\,\text{mA} \quad (\text{at } V_f^{typ})$$

$$I_{LED} = (5.0 - 3.05)/22 = 89\,\text{mA} \quad (\text{at } V_f^{typ} + 0.1)$$

Only a 4.3% variation. The 5 V supply gives much better current stability.

8.2 LDOs and the 5 V Supply Problem

A Linear Dropout Regulator (LDO) produces a regulated output voltage from an input. But it requires the input to be *higher* than the output by at least the **dropout voltage** (V_{DO}):

$$V_{in} \geq V_{out} + V_{DO}$$

Common Mistake

You cannot regulate 5 V from a 5 V supply. Even the best LDOs have $V_{DO} \geq 0.1\,\text{V}$. A 5.0 V-to-5.0 V LDO would require 5.1 V input minimum, and most need 5.5–6.2 V. If your supply is exactly 5 V, connect the LED circuit *directly* to the supply (with bypass capacitors)

rather than through an LDO.

Regulator	V_{DO}	**Min Input for 5 V out**	**Verdict for 5 V supply**
LM7805	∼2.0 V	7.0 V	Cannot use on 5 V supply
LD1117	∼1.2 V	6.2 V	Cannot use on 5 V supply
LM2940	∼0.5 V	5.5 V	Cannot use on 5 V supply
MCP1700	∼0.18 V	5.18 V	Still cannot use on 5 V supply
Direct connection	0	5.0 V	**Correct choice**

Q6. You have a 9 V DC adapter powering an LED circuit. You want to produce a clean 5 V rail for the LEDs.
(a) Can you use an LM7805 (needs $V_{in} \geq 7$ V)?
(b) How much power does the LM7805 dissipate at 400 mAload current?
(c) Would you need a heatsink on the LM7805?
(See Appendix A.)

9 The MOSFET Switch

LEDs in this design are switched on and off by a microcontroller (the Raspberry Pi). Because the GPIO pin can only supply about 16 mA, we use a transistor to switch the much larger LED current.

9.1 Why a MOSFET?

A **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)** is ideal for this application:

- The gate (control input) draws virtually zero current — it only needs a voltage.
- A logic-level N-channel MOSFET turns fully on with a gate voltage of 2–3 V, well within the 3.3 V GPIO output.
- When fully on, the drain-to-source resistance $R_{DS(on)}$ is very low (milliohms to tens of milliohms), so it drops very little voltage.

9.2 The Gate Circuit

Two small resistors protect the MOSFET gate:

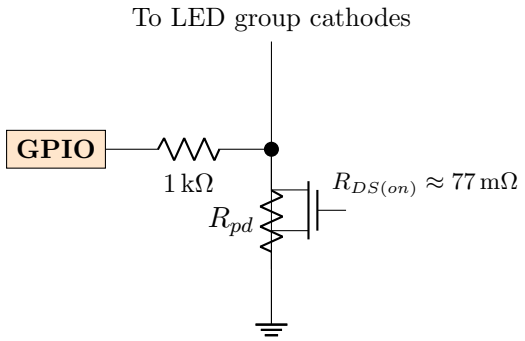


Figure 4: MOSFET gate circuit. R_g limits the current spike when the gate capacitance charges. R_{pd} holds the gate at 0 V when the GPIO pin is not actively driving it.

- R_g **(1 kΩ):** The gate has a small capacitance. Without a series resistor, turning on the MOSFET would demand a large current spike from the GPIO to charge it. 1 kΩ limits this to 3.3 V/1000 Ω = 3.3 mA, well within GPIO capability.
- R_{pd} **(100 kΩ):** During boot, software crashes, or GPIO reconfiguration, the pin may float. Without R_{pd}, the gate could float to any voltage and partially turn on the LEDs at an uncontrolled time. R_{pd} ensures the gate is firmly at 0 V (LED off) unless the GPIO actively drives it high.

Q7. A MOSFET with $R_{DS(on)} = 0.077\,\Omega$ switches a group of 4 LEDs drawing a total of 490 mA.

(a) What voltage drops across the MOSFET when it is on?
(b) How much power does the MOSFET dissipate?
(c) Does it need a heatsink?

(See Appendix A.)

10 Reading the Violet LED Datasheet: A Case Study

The violet LED used in this project (supplier row 1015, 410 nm dominant wavelength) illustrates several important real-world complications.

10.1 What the Datasheet Says

Parameter	Value
Dominant wavelength	410 nm
Forward voltage V_f^{typ}	2.8 V
Forward voltage V_f^{max}	4.0 V
Maximum current I_{max}	30 mA
Maximum power P_{max}	120 mW
Package	0603 SMD

10.2 Design Implications

The V_f spread is very wide: from 2.8 V to 4.0 V — a 1.2 V range. This means different LEDs from the same reel can behave very differently on the same resistor.

The I_{max} is very low: only 30 mA. This is consistent with $P_{max} = V_f^{max} \times I_{max} = 4.0 \times 0.030 = 120\,\text{mW}$. It confirms that the datasheet is internally consistent.

Given: $V_{CC} = 5.0\,\text{V}$, $R = 120\,\Omega$ (chosen to give Medium brightness at typical V_f).
At $V_f^{typ} = 2.8\,\text{V}$:

$$I_{LED} = \frac{5.0 - 2.8}{120} = \frac{2.2}{120} = 18.3\,\text{mA}$$

$$P_{LED} = 2.8 \times 18.3 = 51.2\,\text{mW} \quad (< 120\,\text{mW}\checkmark)$$

$$\%I_{max} = 18.3/30 = 61\% \quad (\text{Medium level}\checkmark)$$

At $V_f^{max} = 4.0\,\text{V}$:

$$I_{LED} = \frac{5.0 - 4.0}{120} = \frac{1.0}{120} = 8.3\,\text{mA}$$

$$P_{LED} = 4.0 \times 8.3 = 33.3\,\text{mW} \quad (< 120\,\text{mW}\checkmark)$$

Conclusion: The same 120 Ω resistor gives 8.3 mAto 18.3 mAdepending on the individual LED's V_f. This is a 2.2× variation in brightness. For a measurement application, this is acceptable because the camera exposure time compensates for group-level brightness; within the group, brighter LEDs simply have a lower V_f.
Why 6 LEDs instead of 4? At only 18 mAeach, a single 0603 violet LED is much dimmer than one power LED. Using 6 multiplies total brightness by 1.5× at minimal cost, while keeping each LED safely within its 30 mAlimit. Total group current = $6 \times 18.3 = 110\,\text{mA}$. Well within budget.

11 Putting It All Together: Design Checklist

- ☐ **Choose LEDs** matching wavelength, package, and price goals.
- ☐ **Record** V_f^{typ}, V_f^{min}, V_f^{max}, I_{max}, P_{max} from the datasheet.
- ☐ **Check supply voltage headroom:** $V_R = V_{CC} - V_f^{typ} \geq 1\,\text{V}$.
- ☐ **Decide number of LEDs** N per group.
- ☐ **Select operating current** I_{LED} (target 50–75% of I_{max}).
- ☐ **Check power budget:** $N \times I_{LED} \leq I_{budget}$. Reduce I_{LED} if necessary.
- ☐ **Calculate** $R_{calc} = (V_{CC} - V_f^{typ})/I_{LED}$.
- ☐ **Round to nearest E24 value** R_{E24}.
- ☐ **Calculate actual current** $I_{LED}^{act} = (V_{CC} - V_f^{typ})/R_{E24}$.
- ☐ **Verify** $I_{LED}^{act} < I_{max}$ and $N \times I_{LED}^{act} < I_{budget}$.
- ☐ **Calculate resistor power** $P_R = (I_{LED}^{act})^2 \times R_{E24}$.
- ☐ **Select resistor package** rated for at least $2 \times P_R$.
- ☐ **Check V_f^{max} scenario:** verify LED still lights at minimum current.
- ☐ **Select MOSFET** with $I_D > N \times I_{LED}^{act}$, logic-level gate threshold $< 3.3\,\text{V}$.
- ☐ **Add gate resistor** (1 kΩ) and pull-down (100 kΩ).
- ☐ **Add bypass capacitors** (100 µF bulk + 100 nF ceramic at supply input; 10 µF near each LED group).
- ☐ **Add polyfuse** (hold current ≥ max group current, trip ≤ 1 A).

12 Comprehensive Exercises

Work through each exercise completely, showing all steps. Check your answers in Appendix A.

Exercise 1. Design a circuit for a single blue LED on a 3.3 V supply.
LED specs: $V_f^{typ} = 2.75\,\text{V}$, $V_f^{max} = 3.1\,\text{V}$, $I_{max} = 30\,\text{mA}$.
Target: Medium brightness.

(a) What is I_{LED}^{target} at Medium level?
(b) What is V_R using V_f^{typ}?
(c) Is the headroom (V_R) adequate? (Is it ≥1 V?)
(d) Calculate R_{calc}.
(e) What is the nearest E24 value?
(f) What is I_{LED}^{actual}?
(g) What is the resistor power P_R? What package do you need?
(h) What is I_{LED} at V_f^{max}? Does the LED still light?

Exercise 2. You have four red LEDs ($V_f = 2.2\,\text{V}$, $I_{max} = 250\,\text{mA}$) on a 5 V supply. Your current budget is 400 mA(not 500 mA— this supply is more constrained).

(a) What is the maximum allowed current per LED?
(b) What percentage of I_{max} is that?
(c) Calculate R_{calc} for this current.
(d) Round to E24. What is I_{LED}^{actual} and I_{total}?
(e) Calculate P_R. What package?
(f) Now recalculate for Low and High brightness. Can you reach High within budget?

Exercise 3. A green LED has $V_f^{typ} = 2.80\,\text{V}$ and $V_f^{max} = 3.20\,\text{V}$. You choose $R = 18\,\Omega$ on a 5.0 V supply.

(a) What is I_{LED} at V_f^{typ}?
(b) What is I_{LED} at V_f^{max}?
(c) What is the ratio $I(V_f^{max})/I(V_f^{typ})$ expressed as a percentage?
(d) Is the headroom (V_R at V_f^{max}) at least 1 V?
(e) Repeat for a supply of 3.3 V with $R = 2\,\Omega$. What do you notice?

Exercise 4. Design resistors for all four channels of a color analyzer. Supply: 5.0 V, budget: 500 mAper group. Use Medium brightness (but cap at budget limit). LEDs:

Channel	V_f^{typ}	I_{max}	N LEDs	Notes
Violet	2.80 V	30 mA	6	Estimated I_{max}
Blue	2.95 V	240 mA	4	
Green	2.80 V	240 mA	4	
Red	2.20 V	250 mA	4	

For each channel: (a) target current per LED, (b) R_{calc}, (c) R_{E24}, (d) I_{LED}^{actual}, (e) I_{total}, (f) P_R and minimum package.

Exercise 5. For each scenario, state whether the proposed resistor package is adequate, marginal, or dangerously undersized, and explain why.

(a) $I_{LED} = 20\,\text{mA}$, $R = 100\,\Omega$, package = 0402 (63 mWrating)
(b) $I_{LED} = 120\,\text{mA}$, $R = 18\,\Omega$, package = 0603 (100 mWrating)
(c) $I_{LED} = 115\,\text{mA}$, $R = 24\,\Omega$, package = 1206 (250 mWrating)
(d) $I_{LED} = 80\,\text{mA}$, $R = 20\,\Omega$, package = 0805 (125 mWrating)

A Answer Key

Quick Check 2.1

Blue LED, $V_f = 2.95\,\text{V}$, $V_{CC} = 5\,\text{V}$, target $I_{LED} = 80\,\text{mA}$.

$$V_R = 5.0 - 2.95 = 2.05\,\text{V}$$

$$R_{calc} = \frac{2.05}{0.080} = 25.6\,\Omega$$

Nearest E24: $25.6\,\Omega \rightarrow$ divide by 10 $\rightarrow 2.56 \rightarrow$ nearest E24 values are 2.4 and 2.7. Distance: $|2.56 - 2.4| = 0.16$; $|2.56 - 2.7| = 0.14$. Nearest is **2.7** $\rightarrow$ **27** Ω.
Actual current: $I_{LED} = 2.05/27 = 75.9\,\text{mA}$(slightly under target, fine).

Quick Check 3.1

$V_f = 2.8\,\text{V}$, $I_{max} = 240\,\text{mA}$, $P_{max} = 800\,\text{mW}$, $I = 150\,\text{mA}$.
(a) $P_{LED} = 2.8 \times 0.150 = 0.42\,\text{W} = 420\,\text{mW}$.
(b) $420\,\text{mW} < 800\,\text{mW}$ — within limit. ✓
(c) $150/240 = 62.5\%$ of I_{max}.

Quick Check 4.1

Red LED, $I_{max} = 250\,\text{mA}$, 4 LEDs, 500 mAbudget.
(a) $I_{Medium} = 0.60 \times 250 = 150\,\text{mA}$per LED.
(b) $I_{total} = 4 \times 150 = 600\,\text{mA}$. **Exceeds 500 mAbudget.**
(c) $I_{max,LED} = 500/4 = 125\,\text{mA}$per LED.
(d) $125/250 = 50\%$ of I_{max}. Still a reasonable operating point.

Quick Check 6.1

(a) Blue: $P_R = (0.114)^2 \times 18 = 0.01300 \times 18 = 234\,\text{mW}$. Need 1206 (250 mW) minimum; 2× derating suggests 2512 (1 W).
(b) Violet: $P_R = (0.018)^2 \times 120 = 0.000324 \times 120 = 38.9\,\text{mW}$. 0402 (63 mW) is adequate; 0603 comfortable.
(c) Red: $P_R = (0.117)^2 \times 24 = 0.01369 \times 24 = 328\,\text{mW}$. Exceeds 1206 (250 mW) rating — use **2512 (1 W)** or use $R = 27\,\Omega$ to reduce current slightly.

Quick Check 7.1

Blue LEDs, $V_f = 2.95\,\text{V}$, $R_i = 18\,\Omega$, $V_{CC} = 5.0\,\text{V}$, $N = 4$.
(a) $I_{LED} = (5.0 - 2.95)/18 = 2.05/18 = 113.9\,\text{mA}$.
(b) $I_{total} = 4 \times 113.9 = 455.6\,\text{mA}$.
(c) With $R_{sh} = 5\,\Omega$: $I_{LED} = (5.0 - 2.95)/(18 + 5) = 2.05/23 = 89.1\,\text{mA}$.
(d) $P_{Rsh} = (0.4556)^2 \times 5 = 0.2076 \times 5 = 1.04\,\text{W}$. Must use a wirewound resistor rated $\geq 2\,\text{W}$.

Quick Check 8.1

9 V supply, LM7805 (needs $V_{in} \geq 7\,\text{V}$), 400 mAload.
(a) Yes. $9\,\text{V} \geq 7\,\text{V}$. The LM7805 **can** be used.
(b) Power dissipated in LM7805 $= (V_{in} - V_{out}) \times I = (9 - 5) \times 0.4 = 4 \times 0.4 = 1.6\,\text{W}$.
(c) **Yes, a heatsink is required.** The LM7805 in TO-220 can only dissipate about 1 W without a heatsink in free air. At 1.6 W it will overheat and go into thermal shutdown. Use a clip-on TO-220 heatsink.

Quick Check 9.1

IRL540N MOSFET, $R_{DS(on)} = 0.077\,\Omega$, $I = 490\,\text{mA}$.
(a) $V_{DS} = I \times R_{DS(on)} = 0.490 \times 0.077 = 37.7\,\text{mV}$. Negligible.
(b) $P_{FET} = I^2 \times R_{DS(on)} = (0.490)^2 \times 0.077 = 0.240 \times 0.077 = 18.5\,\text{mW}$.
(c) No heatsink needed. 18.5 mWis far below the TO-220 package's thermal limit. The device will barely warm up.

Exercise 12.1

Blue LED, $V_f^{typ} = 2.75\,\text{V}$, $V_f^{max} = 3.1\,\text{V}$, $I_{max} = 30\,\text{mA}$, $V_{CC} = 3.3\,\text{V}$.
(a) $I_{Medium} = 0.60 \times 30 = 18\,\text{mA}$.
(b) $V_R = 3.3 - 2.75 = 0.55\,\text{V}$.
(c) **Headroom is inadequate.** $0.55\,\text{V} < 1\,\text{V}$. Current will be very sensitive to V_f variation. This supply-LED combination is marginal at best.
(d) $R_{calc} = 0.55/0.018 = 30.6\,\Omega$.
(e) Nearest E24: **30** Ω.
(f) $I_{LED} = 0.55/30 = 18.3\,\text{mA}$.
(g) $P_R = (0.0183)^2 \times 30 = 0.000335 \times 30 = 10.0\,\text{mW}$. 0402 is fine (63 mWrated).
(h) At $V_f^{max} = 3.1\,\text{V}$: $V_R = 3.3 - 3.1 = 0.2\,\text{V}$. $I_{LED} = 0.2/30 = 6.7\,\text{mA}$. The LED is very dim but still lit. *Key lesson:* a 3.3 V supply is too close to the V_f of blue LEDs for stable, repeatable results. Use 5 V if possible.

Exercise 12.2

Red LED, $V_f = 2.2\,\text{V}$, $I_{max} = 250\,\text{mA}$, $N = 4$, $V_{CC} = 5\,\text{V}$, budget=400 mA.
(a) $I_{max/LED} = 400/4 = 100\,\text{mA}$.
(b) $100/250 = 40\%$ of I_{max}. This is the Low level.
(c) $R_{calc} = (5.0 - 2.2)/0.100 = 2.8/0.100 = 28\,\Omega$.
(d) Nearest E24: **27** Ω. $I_{LED} = 2.8/27 = 103.7\,\text{mA}$. $I_{total} = 4 \times 103.7 = 414.8\,\text{mA}$. Slightly over 400 mA; use **30** Ω instead. $I_{LED} = 2.8/30 = 93.3\,\text{mA}$. $I_{total} = 373.3\,\text{mA}$. ✓
(e) $P_R = (0.0933)^2 \times 30 = 261\,\text{mW}$. Use 1206 (250 mW) or larger.
(f) Low with 30 Ω: done above.
Medium (60%): $I_{target} = 150\,\text{mA}$; $I_{total} = 600\,\text{mA}$— **exceeds 400 mAbudget**. Medium must be capped at 100 mA(same as Low for this budget).
High (80%): $I_{target} = 200\,\text{mA}$; $I_{total} = 800\,\text{mA}$— far over budget. **High is not achievable within a 400 mAbudget with 4 LEDs at $I_{max} = 250\,\text{mA}$.** The budget limits this design to approximately the Low operating level.

Exercise 12.3

Green LED, $V_f^{typ} = 2.80\,\text{V}$, $V_f^{max} = 3.20\,\text{V}$, $R = 18\,\Omega$, $V_{CC} = 5.0\,\text{V}$.
(a) $I_{LED}(V_f^{typ}) = (5.0 - 2.80)/18 = 2.20/18 = 122.2\,\text{mA}$.
(b) $I_{LED}(V_f^{max}) = (5.0 - 3.20)/18 = 1.80/18 = 100.0\,\text{mA}$.
(c) $100.0/122.2 = 81.8\%$. The dimmest LED produces 82% of the brightest LED's output — an 18% variation. Acceptable for most applications.
(d) V_R at $V_f^{max} = 5.0 - 3.2 = 1.8\,\text{V} > 1\,\text{V}$. ✓Adequate headroom.
(e) At 3.3 V with $R = 2\,\Omega$: $I_{LED}(V_f^{typ}) = (3.3 - 2.8)/2 = 0.5/2 = 250\,\text{mA}$. Exceeds I_{max}! $I_{LED}(V_f^{max}) = (3.3 - 3.2)/2 = 0.1/2 = 50\,\text{mA}$. The ratio is $50/250 = 20\%$: a 5× variation. **The low-voltage supply makes the circuit extremely sensitive to V_f variation.**

Exercise 12.4

Ch.	I_{target}	R_{calc}	R_{E24}	I_{act}	I_{tot}	P_R	**Package**
Violet	18.0 mA	122.2 Ω	120 Ω	18.3 mA	110 mA	39.9 mW	0402
Blue	114 mA*	18.0 Ω	18 Ω	113.9 mA	455 mA	234 mW	1206
Green	122 mA*	18.0 Ω	18 Ω	122.2 mA	489 mA	268 mW	1206
Red	117 mA*	23.9 Ω	24 Ω	116.7 mA	467 mA	327 mW	2512

*Budget-constrained: 500 mA/4 = 125 mA max; round to nearest E24 that keeps $I_{total} \leq$ 500 mA. Violet: $I_{max} = 30\,\text{mA}$, Medium = $0.60 \times 30 = 18\,\text{mA}$; 6 LEDs ×18.3 = 110 mA. Red: 22 Ωgives $I_{total} = 509\,\text{mA}$(over); use 24 Ω.

Exercise 12.5

(a) $P_R = (0.020)^2 \times 100 = 40\,\text{mW}$. 0402 rated 63 mW: $40 < 63$. **Adequate**, though 2× derating suggests 0603 for comfort.
(b) $P_R = (0.120)^2 \times 18 = 259\,\text{mW}$. 0603 rated 100 mW: $259 \gg 100$. **Dangerously undersized.** The resistor will overheat and fail. Use 1206 at minimum.
(c) $P_R = (0.115)^2 \times 24 = 317\,\text{mW}$. 1206 rated 250 mW: $317 > 250$. **Marginal to undersized.** Exceeds the 1206 rating. Use 2512 (1 W) for 2× derating. At minimum, increase R to reduce current slightly.
(d) $P_R = (0.080)^2 \times 20 = 128\,\text{mW}$. 0805 rated 125 mW: $128 \approx 125$. **Marginal.** Barely at the limit; 2× derating is violated. Use 1206 for safety.